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United States Patent	12410772
Kind Code	B2
Date of Patent	September 09, 2025
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Ultracapacitor pitch energy module

Abstract

A pitch energy module comprising one or more ultracapacitors storing electrical energy for a wind turbine emergency pitch energy event. The pitch energy module replaces at least one battery within a battery housing of a wind turbine and interfaces with the existing battery wiring harness to communicate with a control system of the wind turbine. The pitch energy module is installed without further modification to the battery housing or the battery wiring harness.

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Appl. No.: 18/948700

Filed: November 15, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250067245 A1	Feb. 27, 2025

Related U.S. Application Data

continuation parent-doc US 18184961 20230316 US 12173689 child-doc US 18948700
continuation parent-doc US 17541495 20211203 US 11624348 20230411 child-doc US 18184961
continuation parent-doc US 17338926 20210604 US 11193471 20211207 child-doc US 17541495
continuation parent-doc US 17180329 20210219 US 11073130 20210727 child-doc US 17338926

Publication Classification

Int. Cl.: **F03D7/02** (20060101); **F03D9/11** (20160101); **H02J7/00** (20060101); H02J7/34 (20060101)

U.S. Cl.:

CPC **F03D7/0224** (20130101); **F03D7/0264** (20130101); **F03D9/11** (20160501); **H02J7/0042** (20130101); F05B2240/85 (20200801); F05B2260/42 (20130101); F05B2270/328 (20130101); F05B2270/504 (20130101); H02J7/345 (20130101); H02J2207/50 (20200101)

Field of Classification Search

CPC: F03D (7/0224); F03D (7/0264); H02J (7/0042); H02J (7/345); H02J (2207/50); F05B (2260/42); F05B (2270/328)

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Background/Summary

RELATED APPLICATIONS (1) This patent application is a continuation application claiming priority benefit, with regard to all common subject matter, of U.S. patent application Ser. No. 18/184,961, filed Mar. 16, 2023, and entitled “ULTRACAPACITOR PITCH ENERGY MODULE” (“the '961 application”). The '961 application is a continuation application claiming priority benefit, with regard to all common subject matter, of U.S. patent application Ser. No. 17/541,495, filed Dec. 3, 2021, now U.S. Pat. No. 11,624,348, issued Apr. 11, 2023, and entitled “ULTRACAPACITOR PITCH ENERGY MODULE” (“the '348 patent”). The '348 patent is a continuation application claiming priority benefit, with regard to all common subject matter, of U.S. patent application Ser. No. 17/338,926, filed Jun. 4, 2021, now U.S. Pat. No. 11,193,471, issued Dec. 7, 2021, and entitled “ULTRACAPACITOR PITCH ENERGY MODULE” (“the '471 patent”). The '471 patent is a continuation application claiming priority benefit, with regard to all common subject matter, of U.S. patent application Ser. No. 17/180,329, filed Feb. 19, 2021, now U.S. Pat. No. 11,073,130, issued Jul. 27, 2021, and entitled “ULTRACAPACITOR PITCH ENERGY MODULE” (“the '130

patent”). The above-referenced application and patents are hereby incorporated by reference in their entirety into the present application.

BACKGROUND

1. Field

(1) Embodiments of the invention relate to pitch energy storage systems for wind turbines. More specifically, embodiments of the invention relate to pitch energy modules employing ultracapacitors for storing electrical energy, wherein the pitch energy module replaces existing batteries within wind turbine pitch energy storage systems.

2. Related Art

(2) Typically, wind turbines store energy for emergency pitch events within batteries mounted within a battery housing disposed within the wind turbine. However, batteries have a range of deficiencies such as slow charging and discharging times, as well as temperature dependence. Additionally, the lifetime of batteries may be limited to a certain number of charge cycles.

(3) Ultracapacitors have been known as an alternative energy storage device for wind turbine emergency pitch energy events. Ultracapacitors have quicker charging and discharging times, operability for a range of temperatures, and suitability for significantly more charging cycles when compared with batteries. However, existing ultracapacitor pitch energy devices are also associated with a number of drawbacks. Typical ultracapacitor pitch energy devices require extensive modifications to the battery housing for installation, such as adapters and additional wiring harnesses, which increases the cost of installation. For example, typical ultracapacitor pitch energy devices cannot interface with the existing battery wiring harness.

(4) Additionally, typical ultracapacitor pitch energy devices distribute a much lower equivalent series resistance when compared with that of batteries. Accordingly, placing ultracapacitor pitch energy devices in series with the existing batteries increases the electrical load on the batteries because of the disparity in the equivalent series resistance. As such, it is problematic if not entirely unfeasible to replace the existing batteries incrementally, i.e., have a mix of batteries and ultracapacitors in the pitch energy system, where the batteries are replaced as they become unusable. Therefore, administrators of wind turbines are frequently forced to replace all batteries with ultracapacitors, even though many of the batteries are still usable, adding increased cost.

(5) Accordingly, what is needed is an ultracapacitor pitch energy module designed to replace at least one battery within a pitch energy system of a wind turbine without requiring additional installation modifications to the battery housing, battery harness, wind turbine control system, or other hardware or software associated with the wind turbine's pitch energy system. Further, the ultracapacitor pitch energy module should have an equivalent series resistance greater than or similar to that of the battery being replaced such that the module can be safely placed in series with existing batteries.

SUMMARY

(6) Embodiments of the invention solve the above-mentioned problems by providing a pitch energy module for replacing at least one battery within an electric pitch energy system of a wind turbine. In some embodiments, the pitch energy module is mounted within a battery housing of the wind turbine and coupled to a battery wiring harness to thereby communicate with a control system of the wind turbine. In some embodiments, the pitch energy module is configured to provide electrical power from one or more ultracapacitors to the wind turbine during an emergency pitch event.

(7) A first embodiment of the invention is directed to a pitch energy module for replacing at least one battery within an electric pitch control system of a wind turbine, the at least one battery mounted in the wind turbine in a battery housing and operably coupled with a control system of the wind turbine via at least a battery wiring harness, the pitch energy module comprising a pitch energy module housing sized for mounting in the battery housing upon replacement of the at least one battery with the pitch energy module, one or more ultracapacitors mounted within the pitch

energy module housing, the one or more ultracapacitors configured to store electrical energy for a wind turbine emergency pitch event, a microprocessor mounted on or in the pitch energy module housing, the microprocessor processing a first set of information associated with the one or more ultracapacitors, a non-volatile memory communicatively coupled with the microprocessor for storing a second set of information associated with the pitch energy module, a communications adapter, mounted on or in the pitch energy module housing and communicatively coupled with the microprocessor, configured to interface with a battery communications cable of the battery wiring harness to thereby transfer at least one signal comprising the first set of information associated with the one or more ultracapacitors to the control system of the wind turbine, and a charger mounted on or in the pitch energy module housing for electrically charging the one or more ultracapacitors.

(8) A second embodiment of the invention is directed to a pitch energy module for replacing a battery within an electric pitch control system of a wind turbine, the battery mounted in the wind turbine in a battery housing and operably coupled with the wind turbine's control system via at least a battery wiring harness, the pitch energy module comprising a pitch energy module housing sized for mounting in the battery housing upon replacement of the battery with the pitch energy module, one or more ultracapacitors mounted within the pitch energy module housing, the one or more ultracapacitors configured to store electrical energy for a wind turbine emergency pitch event, a microprocessor mounted on or in the pitch energy module housing, the microprocessor for processing a first set of information associated with the one or more ultracapacitors, a non-volatile memory communicatively coupled with the microprocessor for storing a second set of information associated with the pitch energy module, a positive terminal electrically coupled to the one or more ultracapacitors, the positive terminal comprising a first bolt fastener, a negative terminal electrically coupled to the one or more ultracapacitors, the negative terminal comprising a second bolt fastener, wherein the battery comprises a first equivalent series resistance, wherein the first bolt fastener and the second bolt fastener achieve a second equivalent series resistance for the one or more ultracapacitors that is higher than the first equivalent series resistance of the battery, a communications adapter, mounted on or in the pitch energy module housing and communicatively coupled with the microprocessor, configured to interface with a battery communications cable of the battery wiring harness to thereby transfer at least one signal comprising the first set of information associated with the one or more ultracapacitors to the control system of the wind turbine, and a charger mounted on or in the pitch energy module housing for electrically charging the one or more ultracapacitors.

(9) A third embodiment of the invention is directed to a pitch energy system for replacing at least one battery within an electric pitch control system of a wind turbine, the at least one battery mounted in the wind turbine in a battery housing and operably coupled with the wind turbine's control system via at least a battery wiring harness, the pitch energy system comprising a plurality of pitch energy modules electrically connected in series, each of the plurality of pitch energy modules comprising a pitch energy module housing sized for mounting in the battery housing upon replacement of the at least one battery, one or more ultracapacitors mounted within the pitch energy module housing, the one or more ultracapacitors configured to store electrical energy for a wind turbine emergency pitch event, a microprocessor mounted on or in the pitch energy module housing, the microprocessor processing a first set of information associated with the one or more ultracapacitors, a non-volatile memory communicatively coupled with the microprocessor for storing a second set of information associated with the pitch energy module, a communications adapter, mounted on or in the pitch energy module housing and communicatively coupled with the microprocessor, configured to interface with a battery communications cable of the battery wiring harness to thereby transfer at least one signal comprising the first set of information associated with the one or more ultracapacitors to the control system of the wind turbine, and a charger mounted on or in the pitch energy module housing for electrically charging the one or more ultracapacitors.

(10) A fourth embodiment of the invention is directed to a method of use of a pitch energy module

for replacing at least one battery in an electric pitch control system of a wind turbine, the method comprising the steps of charging a plurality of ultracapacitors mounted within a pitch energy module housing of the pitch energy module, storing electrical energy within the plurality of ultracapacitors, transmitting a first signal comprising information associated with the plurality of ultracapacitors to a control system of the wind turbine, and supplying electrical energy from the plurality of ultracapacitors to a plurality of motors within the wind turbine during an emergency pitch event to adjust the pitch of the wind turbine blades.

(11) A fifth embodiment of the invention are directed to a method for replacing a battery within an electric pitch control system of a wind turbine with a pitch energy module, the method comprising the steps of disconnecting a battery wiring harness from the battery, removing the battery from a battery housing of the wind turbine, mounting the pitch energy module in the battery housing, and connecting the battery wiring harness to the pitch energy module.

(12) This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

(1) Embodiments of the invention are described in detail below with reference to the attached drawing figures, wherein:

(2) FIG. 1 depicts a wind turbine relating to some embodiments of the invention;

(3) FIG. 2A depicts a battery housing relating to some embodiments of the invention;

(4) FIG. 2B depicts a battery housing relating to some embodiments of the invention;

(5) FIG. 3A depicts an isometric view of a pitch energy module relating to some embodiments of the invention;

(6) FIG. 3B depicts a top view of a pitch energy module relating to some embodiments of the invention;

(7) FIG. 3C depicts a rear view of a pitch energy module relating to some embodiments of the invention;

(8) FIG. 3D depicts a top view of a pitch energy module interfacing with a battery wiring harness relating to some embodiments of the invention;

(9) FIG. 4A depicts an isometric view of a pitch energy module relating to some embodiments of the invention;

(10) FIG. 4B depicts a front view of a pitch energy module relating to some embodiments of the invention;

(11) FIG. 4C depicts a top view of a pitch energy module interfacing with a battery wiring harness relating to some embodiments of the invention;

(12) FIG. 5 depicts a circuit board relating to some embodiments of the invention;

(13) FIG. 6 depicts a control diagram relating to some embodiments of the invention;

(14) FIG. 7 depicts a method for using a pitch energy module relating to some embodiments of the invention; and

(15) FIG. 8 depicts a method for installing a pitch energy module relating to some embodiments of the invention.

(16) The drawing figures do not limit the invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

(17) The following detailed description references the accompanying drawings that illustrate specific embodiments in which the invention can be practiced. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

(18) In this description, references to “one embodiment,” “an embodiment,” or “embodiments” mean that the feature or features being referred to are included in at least one embodiment of the technology. Separate references to “one embodiment,” “an embodiment,” or “embodiments” in this description do not necessarily refer to the same embodiment and are also not mutually exclusive unless so stated and/or except as will be readily apparent to those skilled in the art from the description. For example, a feature, structure, act, etc. described in one embodiment may also be included in other embodiments, but is not necessarily included. Thus, the technology can include a variety of combinations and/or integrations of the embodiments described herein.

(19) Wind turbines harvest renewable energy from the wind using a plurality of blades attached to a rotor to drive rotation of the rotor and convert the energy into electrical energy by turning a generator. In some cases, it may be desirable to adjust the pitch of the blades of the wind turbine. For example, during a wind turbine emergency pitch event, stored electrical energy may be supplied to motors within the wind turbine to disengage the blades. Here, the blades may be rotated at a 90 degree angle such that they are perpendicular to the wind and decrease the rotational velocity of the rotor or prevent the rotor from turning. In some embodiments, it may be desirable to adjust the pitch of the blades when a safety critical fault is experienced by the wind turbine to reduce the rotational velocity of the blades and rotor. Additionally, it may be desirable to adjust the pitch of the blades when a grid event is experienced where power is lost to the entire grid. Here, it is desirable to disengage the blades using backup power before power to the turbine control system is completely lost.

(20) Embodiments of this disclosure provide a pitch energy module for replacing at least one battery within an electric pitch control system of a wind turbine. In some embodiments, the pitch energy module is configured to interface with the existing installation system intended for the battery such that installation time and cost is reduced. Further, embodiments of the invention provide a pitch energy module with an equivalent series resistance that is similar to that of a battery such that the pitch energy module may be used in the pitch control system in tandem with existing batteries. Accordingly, embodiments are contemplated where batteries may be replaced incrementally with pitch energy modules, as needed.

(21) Turning first to FIG. 1, a wind turbine **10** is depicted relating to some embodiments of the invention. In some embodiments, the wind turbine **10** comprises a tower **12** extending vertically and supporting a nacelle **14**. In some embodiments, a generator and a controller of the wind turbine **10** are housed within the nacelle **14**. Additionally, the wind turbine **10** further comprises a rotor **16** rotatably secured to the nacelle **14**. The rotor **16** supports a plurality of blades **18** extending radially outwards from the rotor **16**. For example, in some embodiments, the wind turbine **10** comprises three blades **18**, as shown. In some embodiments, each of the blades **18** is rotatably secured to the rotor **16** via a slew ring bearing.

(22) In some embodiments, a plurality of motors are disposed within the rotor **16** for rotating the blades **18** with respect to the rotor **16** to thereby adjust the pitch of the blades **18**. In some embodiments, it may be desirable to adjust the pitch of the blades **18** during an emergency pitch event of the wind turbine **10** or to test the pitch system of the wind turbine **10**. Here, an emergency pitch event may occur when at least one power source has been cut off from the wind turbine **10**. Additionally, an emergency pitch event may occur when a maximum rotational velocity of the rotor

16 has been exceeded. When an emergency pitch event occurs, energy is employed from an emergency pitch system of the wind turbine **10** to drive the motors and thereby adjust the pitch of the blades **18** such that the blades **18** are positioned perpendicular to the direction of the wind. Accordingly, the blades **18** are disengaged and the rotational velocity of the rotor **16** is reduced. In some embodiments, a large amount of energy may be used over a short period of time to adjust the pitch of the blades **18**. Accordingly, ultracapacitors may be better suited to store and supply said energy when compared with traditional batteries.

(23) Turning now to FIG. 2A, a battery housing **20** is depicted relating to some embodiments of the invention. The battery housing **20** may be disposed within the rotor **16** of the wind turbine **10** where the rotor **16** is attached to each of the blades **18**. Accordingly, a plurality of battery housings **20** may be disposed within a single rotor **16** at the connection point of each respective blade **18**. For example, a wind turbine **10** with three blades **18** may comprise three battery housings **20**. In some embodiments, the battery housing **20** may be secured to an internal wall of the rotor **16** adjacent to the slew ring connecting the rotor **16** to the blade **18**. In some embodiments, the battery housing **20** comprises a frame **22** for securing a plurality of batteries **24**. In some such embodiments, the frame **22** may be composed of stainless steel sheet metal, aluminum sheet metal, hard plastic, fiberglass, or another suitable material.

(24) In some embodiments, each of the batteries **24** may be removably mounted within the frame **22** using bolts or another suitable fastener. Further, the batteries **24** may be coupled to a control system of the wind turbine **10** using a battery wiring harness **26**. Here, each battery wiring harness **26** comprises a plurality of cables electrically connected to ports and terminals on the battery. For example, each battery **24** may comprise a communications port, an external power input port, a positive terminal, and a negative terminal. Accordingly, the battery wiring harness **26** may be operable to transmit communication signals between the battery **24** and the control system of the wind turbine **10**. Further, the positive and negative terminals of the battery may be used to transfer electrical energy stored within the battery **24** to the motors of the wind turbine **10** during an emergency pitch event. In some embodiments, the batteries **24** are electrically connected in series, as shown, with the positive terminal of one battery wired connected to the negative terminal of the next battery using a cable of the battery wiring harness **26**. Further, it should be understood that, in some embodiments, each of the batteries **24** is coupled with an external battery charger for charging the battery **24**. Here, the external battery charger may be disposed within the battery housing **20**.

(25) The battery housing **20**, as shown in FIG. 2A, comprises a pitch energy module **30** mounted within the battery housing **20**. For example, the pitch energy module **30** may be bolted to the frame **22** of the battery housing **20**. In some embodiments, the pitch energy module **30** is coupled to the control system of the wind turbine **10** using the battery wiring harness **26** in a similar fashion to the batteries **24**, as will be discussed in detail below. In some such embodiments, it may be desirable that the pitch energy module **30** is mounted within the battery housing **20** and connected with the battery wiring harness **26** in the same way as the batteries **24**. Accordingly, the pitch energy module **30** may replace one of the batteries **24** and the corresponding battery charger without requiring additional modifications to the battery housing **20** or the battery wiring harness **26**. Similarly, in some embodiments, the pitch energy module **30** is mounted within the battery housing **20** without additional mounting means or structural modification to the battery housing **20**. Further, it should be understood that, in some embodiments, the pitch energy module **30** is operable to interface with the control system of the wind turbine **10** without modifying software of the control system. Accordingly, the pitch energy module **30** may be configured to communicate the same types of signals with the control system as the battery **24** being replaced.

(26) In some embodiments, the pitch energy module **30** replaces a single battery and battery charger. For example, a pitch energy system that previously had six batteries may replace one of the batteries **24** such that there are five batteries and one pitch energy module **30**, as shown.

Alternatively, in some embodiments, the pitch energy module **30** may replace a plurality of batteries within the battery housing **20**. For example, in some embodiments, one pitch energy module **30** may replace two or more batteries **24** within the battery housing **20**. In some embodiments, it may be desirable to replace batteries **24** within the battery housing **20** incrementally such that one battery is replaced at a time while other batteries are not replaced to thereby operate with varying ages and wear of batteries, as will be discussed in further detail below. Accordingly, the pitch energy module **30** may be installed within the battery housing as needed. For example, a battery **24** may be replaced with a pitch energy module **30** when the battery **24** has exceeded its useful lifetime or when the battery **24** has become faulty. As such, embodiments are contemplated where an unstable battery is replaced with the pitch energy module **30** while other stable batteries **24** remain in the battery housing **20**.

(27) In some embodiments, during a pitch energy event, energy stored within the batteries **24** and the pitch energy module **30** is used to drive the motors and thereby pitch the blades **18** of the wind turbine **10**. It should be understood that embodiments are contemplated where any number of batteries and pitch energy modules **30** are mounted within the battery housing **20**.

(28) Turning now to FIG. 2B, the battery housing **20** is depicted with a plurality of pitch energy modules **30** relating to some embodiments of the invention. Here, each battery **24** and battery charger within the battery housing **20** is replaced with a corresponding pitch energy module **30**. Each of the pitch energy modules **30** is mounted within the battery housing **20** and coupled to the control system of the wind turbine **10** via a respective battery wiring harness **26**. One advantageous feature of embodiments of the invention is that the pitch energy module uses the existing battery harness **26** to both structurally and communicatively mount and couple the pitch energy module (and any included components, as discussed below) within the battery housing **20**. As such, no additional communication cables, structural supports, mounting brackets or hardware, or other communication or mounting components—other than the battery wiring harness **26**—is needed to mount the pitch energy module and have it communicate with the control system of the wind turbine. Here, the pitch energy modules **30** may be electrically connected in series, as shown, using cables of the battery wiring harness **26** to wire the positive terminal of one pitch energy module **30** to a negative terminal of the next pitch energy module **30**.

(29) It should be understood that various other configurations within the battery housing **20** are also contemplated. For example, the battery housing **20** may hold six pitch energy modules **30**, as shown, or may hold four pitch energy modules **30**. Further, the mounting positions of the pitch energy modules **30** may be adjusted within the battery housing **20**. For example, the pitch energy modules **30** may be mounted on a top and bottom row within the battery housing **20**, as shown, but may also be mounted on the sides and may be mounted in a different orientation. In some embodiments, the number of pitch energy modules **30** and the mounting positions within the battery housing **20** may be determined based on the type of wind turbine **10**. For example, a smaller wind turbine **10** may require less power to perform a pitch adjustment and only needs four pitch energy modules **30** within the battery housing **20** for each blade **18**.

(30) Turning now to FIG. 3A, an isometric view of a pitch energy module **30** is depicted relating to some embodiments of the invention. Here, the pitch energy module **30** comprises a pitch energy module housing **32**, which may be composed of sheet metal, hard plastic, fiberglass, or another suitable rigid material. In some embodiments, mounting brackets **34** are secured to the edges of the pitch energy module housing **32**. Accordingly, the mounting brackets **34** may be used to mount the pitch energy module **30** to the frame **22** within the battery housing **20**. In some embodiments, the mounting brackets **34** comprise slotted holes configured to receive a bolt to thereby adjustably secure the pitch energy module **30** within the battery housing **20**.

(31) The pitch energy module **30** further comprises a circuit board **36** mounted on the pitch energy module housing **32**. For example, the circuit board **36** may be mounted on top of the pitch energy module **30**, as shown. In some embodiments, it may be desirable to include a protective cover **38**

placed over the circuit board **36** to protect the circuit board **36** from physical damage, dust, and weathering. Here, the protective cover **38** may be composed of fiberglass, glass, hard plastic, or another rigid material. Further, in some embodiments, the protective cover **38** may be transparent such that the circuit board **36** is visible beneath the protective cover **38**. In yet further embodiments, the circuit board **36** may be mounted in an interior of the pitch energy module **30**.

(32) The pitch energy module **30** further comprises a positive terminal **40** and a negative terminal **42** mounted on the circuit board **36**. In some embodiments, the terminals **40** and **42** protrude from the protective cover **38**, as shown, such that the terminals are accessible for connection to the battery wiring harness **26**. In some embodiments, direct current (DC) power is provided at the terminals **40** and **42** of the pitch energy module **30** for driving the motors of the wind turbine **10** to adjust the pitch of the blades **18**. Additionally, in some embodiments, each of the positive terminal **40** and the negative terminal **42** comprises a bolt fastener **44** for electrically and mechanically coupling a respective cable of the battery wiring harness **26** to the terminal. Here, the terminals **40** and **42** may comprise a threaded hole for receiving the bolt fasteners **44**. In some embodiments, using the bolt fasteners **44** increases an equivalent series resistance associated with the pitch energy module **30**, as will be discussed below.

(33) In some embodiments, the pitch energy module **30** further comprises a communications adapter **46** configured to be coupled to a communications cable of the battery wiring harness **26** to thereby transmit communications with the wind turbine **10**. Here, the communications adapter **46** may be operable to both transmit and receive communications signals with the control system of the wind turbine **10**. In some embodiments, the communications adapter **46** comprises a 15-pin connection port, as shown, for receiving/transmitting communications using 15 different wires of the battery wiring harness **26**. Alternatively, in some embodiments, different types of connection ports may be used. Further, in some embodiments, the pitch energy module **30** comprises an external power input **48**. For example, the external power input **48** may be configured to be coupled to a power cable of the battery wiring harness **26** to receive electrical energy for powering the pitch energy module **30**. In some embodiments, the external power input **48** is an alternating current (AC) power input for receiving AC power from an AC source. In some embodiments, the external power input **48** may comprise a 6-pin connection port, as shown.

(34) In some embodiments, the pitch energy module **30** further comprises a built in charger **50** mounted on the pitch energy module housing **32**. In some embodiments, the charger **50** may be mounted on top of the pitch energy module **30** above the protective cover **38**, as shown. Alternatively, in some embodiments, the charger **50** may be mounted on a front facing side of the pitch energy module **30**, as shown in FIGS. **4A** and **4B**. In some embodiments, it may be desirable that the charger **50** is selectably mounted on the pitch energy module **30** in a specific location based on a configuration of the wind turbine **10**. For example, in some embodiments, the charger **50** may be mounted on the front facing side of the pitch energy module **30** such that the pitch energy module **30** fits into a specific location within the battery housing **20**. It should be understood that different types of wind turbines may comprise battery housings with various different shapes and sizes. Accordingly, it may be desirable that the mounting position of the charger **50** on the pitch energy module **30** is adjustably selectable to fit within the battery housing **20**.

(35) In some embodiments, the charger **50** is held at a floating voltage potential of the pitch energy module **30**. Accordingly, the size of the charger **50** may be reduced as compared to a charger requiring its own electrical grounding system and the charger **50** may be disposed on the pitch energy module **30**. Here, the charger **50** may not comprise an electrical ground to earth but has a floating ground. Thus, the charger **50** may be grounded with respect to the pitch energy module **30**.

(36) In some embodiments, the pitch energy module **30** further comprises a microprocessor **52** mounted on the circuit board **36** for processing information associated with the pitch energy module **30** such as, for example, information relating to one or more ultracapacitors mounted within the pitch energy module **30**.

(37) Turning now to FIG. 3B, a top view of the pitch energy module **30** is depicted relating to some embodiments of the invention. In some embodiments, the charger **50** comprises an additional external power input **48**. Here, it may be desirable to use the external power input **48** mounted on the charger **50** and not the external power input **48** mounted on the circuit board **36**. Further, in some embodiments, the charger **50** may comprise a charger circuit board which may be communicably coupled with the circuit board **36** of the pitch energy module **30**. Additionally, in some embodiments, the charger **50** is mounted on top of the pitch energy module **30**, as shown.

(38) Turning now to FIG. 3C, a rear view of the pitch energy module **30** is depicted relating to some embodiments of the invention. Here, the pitch energy module **30** may be opened such that the internals of the pitch energy module housing **32** are visible. Accordingly, a second internal circuit board **54** is visible mounted at the bottom of the pitch energy module **30** inside the pitch energy module housing **32**. In some embodiments, the second internal circuit board **54** may be mounted within the pitch energy module housing **32** spaced from the bottom using spacers **56**, as shown. Additionally, in some embodiments, spacers **56** may be disposed between the circuit board **36** and the protective cover **38** to create a gap between the circuit board **36** and the protective cover **38**. In some embodiments, it may be desirable to include a gap between the circuit board **36** and the cover **38** to permit airflow to thereby reduce heating on the circuit board **36** and to allow space for components to be mounted on the circuit board **36**.

(39) In some embodiments, the pitch energy module **30** comprises at least one ultracapacitor **60** disposed within the pitch energy module housing **32**, as shown. In FIG. 3C three ultracapacitors **60** are shown, though it should be understood that additional ultracapacitors may be included which are not visible. For example, in some embodiments, six ultracapacitors **60** may be disposed in each pitch energy module **30**. Alternatively, in some embodiments, a single ultracapacitor **60** may be disposed within each pitch energy module **30**. The ultracapacitors **60** are configured to store electrical energy to be used during an emergency pitch event of the wind turbine **10**. Here, stored energy within the ultracapacitors **60** is transmitted to the motors of the wind turbine **10** through the connection with the battery wiring harness **26**. In some embodiments, each of the ultracapacitors **60** comprises one of an electrostatic double-layer capacitor, an electrochemical pseudocapacitor, or a hybrid capacitor. Here, a hybrid capacitor may utilize both electrostatic and electrochemical energy storage techniques.

(40) In some embodiments, it may be desirable to store electrical energy within the ultracapacitors **60** such that a large amount of energy is quickly available. Further, the ultracapacitors **60** of the pitch energy modules **30** may be better suited for periodically storing and releasing large amounts of energy used for the emergency pitch events when compared with the batteries **24**.

(41) In some embodiments, the positive terminal **40** and the negative terminal **42** protrude from the protective cover **38**, as shown. Here, spaces may be cut out of the protective cover **38** such that the terminals **40** and **42** are accessible. Further, the terminals **40** and **42** comprise the bolt fasteners **44**, as described above. The bolt fasteners **44** may be used to secure a cable from the battery wiring harness **26** to each of the terminals **40** and **42**. In some embodiments, the terminals **40** and **42** are electrically coupled to the ultracapacitors **60**. For example, the ultracapacitors **60** may be electrically connected in series. Here, the terminals of each ultracapacitor **60** may be electrically coupled to one of the circuit board **36** or the second internal circuit board **54**. Here, the plurality of ultracapacitors **60** may be sandwiched between the circuit boards **36** and **54**, as shown.

(42) In some embodiments, the pitch energy module **30** comprises at least three mounting brackets **34**, as shown. Here, the mounting brackets **34** may be positioned on the pitch energy module housing **32** corresponding to mounting brackets of a battery **24** being replaced such that the same bolts and mounting holes of the battery housing **20** may be used to mount the pitch energy module **30** in place of the battery **24**. Accordingly, the pitch energy module **30** may replace the battery **24** without requiring additional mounting accommodations. Further, the mounting brackets **34** may be staggered on each side of the pitch energy module housing **32**. For example, the mounting brackets

34 on one side of the pitch energy module housing **32** may be positioned differently than the mounting bracket **34** on the other side, as shown. Accordingly, space is saved within the battery housing **20** because the mounting brackets **34** may be placed in line with mounting brackets of another pitch energy module without interfering.

(43) Turning now to FIG. 3D, a top view of the pitch energy module **30** interfacing with the battery wiring harness **26** is depicted relating to some embodiments of the invention. Here, the battery wiring harness **26** may comprise a plurality of cables configured to connect to various ports and terminals of the battery **24**. In some embodiments, the battery wiring harness **26** comprises a positive terminal cable **41** configured to connect to a positive terminal of the battery **24**, a negative terminal cable **43** configured to connect to a negative terminal of the battery **24**, a communications cable **47** configured to connect to a communications adapter of the battery **24**, and an AC input cable **49** configured to connect to an external power input of the battery **24**. In some embodiments, each of the positive terminal cable **41** and the negative terminal cable **43** comprises a jumper wire. For example, a 6 AWG wire may be used to electrically couple to the terminals.

(44) In some embodiments, the battery **24** is replaced with the pitch energy module **30**, as shown. Here, the positive terminal cable **41** is connected to the positive terminal **40** of the pitch energy module **30**. For example, the positive terminal cable **41** may be secured to the positive terminal **40** using the bolt fastener **44**. In some embodiments, the positive terminal cable **41** connects to a negative terminal of another pitch energy module **30** such that the pitch energy modules are electrically connected in series. Alternatively, the positive terminal cable **41** may be electrically grounded at the other end of the cable. Similarly, the negative terminal cable **43** is connected to the negative terminal **42** of the pitch energy module **30**, as shown, using the bolt fastener **44**. At the other end of the negative terminal cable **43** the cable may be connected to a positive terminal of another pitch energy module **30** or to a power source of the wind turbine **10**. Additionally, in some embodiments, the communications cable **47** may be plugged into the communications adapter **46** of the pitch energy module **30**. Here, in some embodiments, the communications cable **47** may comprise a 15-pin connector for interfacing with the communications adapter **46**. Similarly, the AC input cable **49** may be connected to the external power input **48** of the pitch energy module **30**. Alternatively, in some embodiments, the AC input cable **49** may be plugged into the external power input **48** of the charger **50**, as shown, depending on the configuration of the battery housing **20**. For example, in some embodiments, when the pitch energy module **30** comprises a top mounted charger **50**, the external power input **48** of the charger **50** is used instead of the external power input **48** of the pitch energy module **30**. Accordingly, the pitch energy module **30** is attached to the battery wiring harness **26** of the battery housing **20** using cables **41**, **43**, **47**, and **49** intended to be connected to the battery **24**.

(45) In some embodiments, the pitch energy module **30** comprises an AC input voltage within a range of 85-305 Volts. Further, in some embodiments, the pitch energy module **30** comprises an AC input protection device such as a fuse rated for 2.5 Amps at 250 Volts. In some embodiments, the charge voltage may be provided by the charger **50** at around 14 Volts DC. However, it should be understood that a variety of different voltage and current parameters are contemplated for the pitch energy module **30**. In some embodiments, the pitch energy module **30** may be fully charged in under 35 minutes using the charger **50**. Additionally, in some embodiments, the DC output voltage of the pitch energy module **30** may be adjustably selected between 12 Volts and 16 Volts. In some embodiments, the output voltage may be adjusted by 0.5 Volt increments.

(46) In some embodiments, it may be desirable to adjust the voltage set-point of the pitch energy module **30** to increase the lifetime of the ultracapacitors **60**. Accordingly, it may be desirable to allow an operator to selectably adjust the operating voltage of the pitch energy module **30**. Here, the voltage set-point may be adjusted by one of a manual voltage adjustment device, such as a potentiometer, or a software command. It should be understood, that in some cases the capacity of the ultracapacitors drops over time. As the drop in capacity becomes noticeable, it may be desirable

to increase the voltage set-point to increase the electrical capacity to thereby increase the lifetime of the ultracapacitors **60**. Additionally, in some embodiments, the pitch energy module **30** comprises a wider range of operating temperatures when compared to the battery **24**. For example, the pitch energy module **30** may comprise a rated operating temperature range between -40 to 65° C. Accordingly, the pitch energy module **30** is better suited for a variety of environments where wind turbines may be installed.

(47) Turning now to FIG. **4A**, a pitch energy module **30** with a front mounted charger **50** is depicted relating to some embodiments of the invention. Here, the charger **50** is mounted on the front facing side of the pitch energy module housing **32**, as shown. In some embodiments, this mounting configuration may be desirable to allow the pitch energy module **30** to fit within the battery housing **20** depending on the configuration of the wind turbine **10**. In some embodiments, when the front mounted charger configuration is used, the external power input **48** mounted on the circuit board **36** is used instead of the external power input **48** of the charger **50**. Similarly, FIG. **4B** shows a front view of the pitch energy module **30** with the front mounted charger **50** relating to some embodiments of the invention. It should be understood that a variety of mounting orientations for the pitch energy module **30** are contemplated based on the mounting orientation of the batteries being replaced. For example, in some embodiments, the batteries **24** may be mounted on their side within the battery housing **20**. Accordingly, the pitch energy module **30** may be mounted in a similar orientation within the battery housing **20**.

(48) Turning now to FIG. **4C**, a top view of the pitch energy module **30** with a front mounted charger **50** interfacing with the battery wiring harness **26** is depicted relating to some embodiments of the invention. Here, the cables of the battery wiring harness **26** may be coupled to the pitch energy module **30** similar to as shown in FIG. **3D**, with the positive terminal cable **41** coupled to the positive terminal **40**, the negative terminal cable **43** coupled to the negative terminal **42**, and the communications cable **47** coupled to the communications adapter **46**. However, since the pitch energy module **30** comprises a charger **50** mounted on the front of the module, the AC power input cable **49** is coupled to the external power input **48** of the pitch energy module **30**. Here, the external power input **48** of the charger **50** may not be used based on the configuration of the pitch energy module **30**.

EQUIVALENT SERIES RESISTANCE

(49) In the prior art, batteries within the pitch energy system of a wind turbine are typically placed in series such that the voltages of the batteries are additive. In such prior art systems, each battery is associated with an equivalent series resistance (ESR) based on the internal resistance of the battery and the connections to the terminals of the battery. In some prior art systems, modules including ultracapacitors are installed within the pitch energy system. Typically, such modules comprise a much lower ESR compared to that of the battery being replaced because the terminals of the module are typically welded.

(50) It should be understood that the ultracapacitors within the pitch energy module are not ideal capacitors and therefore, comprise an internal resistance affecting the flow of electrical current within the ultracapacitors. The internal resistance of the ultracapacitors along with the internal resistance associated with the connection means at the positive and negative terminals factor into the overall ESR of the pitch energy module, as discussed in detail below.

(51) Table 1 depicted below shows the estimated ESR given in milliohms for various energy storage devices, such as battery **24**, a pitch energy module **30** with bolted terminals, and a pitch energy module with welded terminals.

(52) TABLE-US-00001 TABLE 1 Equivalent Series Resistance Battery 9 m Ω Welded Terminals 4 m Ω Bolted Terminals 16 m Ω

(53) As discussed above, the terminals of prior art pitch energy modules are typically welded, which decreases the ESR of the pitch energy module. As shown in Table 1, a pitch energy module with welded terminals may comprise an ESR around 4 milliohms. Here, the decreased ESR

associated with the welded terminals may increase the electrical efficiency of the module by allowing current to flow freely from the terminals. However, the reduced ESR may undesirably prevent the pitch energy module **30** from safely being placed in series with the battery **24**. For example, it should be understood that a high ESR device placed in series with a plurality of lower ESR devices may experience a majority of the electrical load. Here, the ESR of the welded terminal pitch energy module may be substantially lower than that of the battery **24** increasing the electrical load on the battery **24**. Because the electrical load of the battery **24** is increased, the battery **24** will be depleted much faster or a potential electrical fault may be experienced. Additionally, the lower ESR associated with welded terminals may undesirably increase the fault current of the module by allowing current to flow freely.

(54) As such, it is desirable that the terminals **40** and **42** of the pitch energy module **30** comprise the bolt fasteners **44** or otherwise distribute an ESR that is similar to or greater than that of the battery **24**. For example, in some embodiments, the pitch energy module **30** with the bolt fasteners **44** comprises an ESR of around 16 milliohms, as shown in Table 1. Accordingly, the batteries **24** and corresponding battery chargers may be replaced with pitch energy modules **30** one by one without damaging the existing batteries. For example, a single battery **24** may be replaced with a pitch energy module **30** such that the pitch energy module **30** is placed in series with the remaining batteries **24**, as shown in FIG. 2A. As such, the installation cost associated with the pitch energy module **30** is greatly reduced because the batteries do not have to be replaced all at once.

(55) In some embodiments, it may be desirable that the ESR is relatively high compared to typical ultracapacitor modules. Here, the ESR may increase the safety of the pitch energy module **30** by decreasing the fault current associated with the pitch energy module **30**. It should be understood that the fault current of the system varies inversely with the ESR because the ESR inhibits the flow of current within the pitch energy module **30**, thereby reducing the fault current. Accordingly, a pitch energy module with a higher ESR will have a lower fault current, thereby improving the safety of the module.

(56) Further, it may be desirable that the ESR of the pitch energy module **30** is similar to or greater than that of the battery **24** such that the pitch energy module **30** can safely be placed in series with the battery **24** without increasing the electrical load experienced by the battery **24**. Accordingly, the bolt fasteners **44** may be used at the terminals **40** and **42** to increase the ESR of the pitch energy module **30**. Here, the bolt fasteners **44** may comprise a sufficient length and electrical resistance to increase the ESR. Accordingly, it may be desirable that the bolt fasteners are composed of a relatively non-conductive material, such as for example, stainless steel or aluminum. Additionally, the length of the bolt fasteners **44** may be increased such that the distance the current must travel is increased, thereby increasing the electrical resistance associated with the connection to the terminals **40** and **42**. As should be appreciated, it may be counterintuitive to increase the resistance of the pitch energy module **30**, because the increased ESR reduces the electrical efficiency of the system. However, embodiments of the present invention recognize the benefits of increasing the ESR of the pitch energy module **30** such that the pitch energy module **30** may be safely placed in series with existing batteries **24**.

(57) Turning now to FIG. 5, the circuit board **36** is depicted relating to some embodiments of the invention. In such embodiments, the circuit board **36** comprises a plurality of circuit traces **62** for electrically connecting the terminals of the plurality of ultracapacitors in series. In some embodiments, the circuit traces **62** are high current circuit traces for transmitting electrical power. Here, the circuit traces **62** may be positioned on the circuit board **36** such that they contact terminals of the ultracapacitors **60** or may otherwise be integrated in or formed into the circuit board **36**. Accordingly, the negative terminal of each ultracapacitor may be coupled to the positive terminal of the next ultracapacitor. Similarly, the second internal circuit board **54** may also comprise circuit traces **62** for electrically connecting the ultracapacitors **60** at the bottom of the pitch energy module housing **32**.

(58) Prior art capacitor banks typically use bus bars to electrically connect capacitors instead of circuit traces **62**. Here, the bus bars are designed to withstand extreme thermal stress associated with extended use of the capacitors. However, in embodiments of the invention, the ultracapacitors **60** are only used for short bursts of energy to adjust the pitch of the blades **18**. The operation time of the ultracapacitors **60** associated with adjusting the pitch of the blades **18** is relatively low when compared with other applications of capacitors. Embodiments of the invention recognize the thermal stress associated with the pitch event does not produce enough heat to significantly raise the temperature of the circuit board **36**. Accordingly, the circuit traces **62** may be used to safely connect the ultracapacitors **60** in series and deliver high-power electrical energy over a short period of time without damaging the circuit board **36**.

(59) The circuit board **36** further comprises pins for the communications adapter **46** and the external power input **48**, as shown. Accordingly, signals may be communicated to either of the communications adapter **46** and the external power input **48** using signal traces on the circuit board **36**.

(60) Turning now to FIG. **6**, an exemplary diagram of a pitch energy control system **600** is depicted relating to some embodiments of the invention. In some embodiments, the pitch energy control system **600** comprises the ultracapacitors **60** electrically coupled to the circuit board **36**. Mounted on the circuit board **36** are the communications adapter **46**, the external power input **48**, the microprocessor **52**, and a non-volatile memory **64**. The non-volatile memory **64** may be communicatively coupled to the microprocessor **52** and comprise an erasable programmable read-only memory (EPROM) for storing information associated with the pitch energy module **30**. In some embodiments, the non-volatile memory **64** stores information relating to the ultracapacitors **60** such as, for example, charge over time.

(61) In some embodiments, the non-volatile memory **64** stores information relating the pitch energy module, such as a number of emergency pitch events, a number of full charges of the ultracapacitors **60** within the pitch energy module **30**, and a number of days of operation of the pitch energy module **30**. However, it should be understood that in some embodiments, the non-volatile memory **64** stores additional information relating to historic operational data of the pitch energy module **30**. In some embodiments, it may be desirable to store such information within the non-volatile memory **64** to monitor and troubleshoot the pitch energy module **30**.

(62) The pitch energy control system **600** further comprises the charger **50** which may be electrically and communicatively connected through the circuit board **36** to the ultracapacitors **60**. Alternatively, in some embodiments, the charger **50** may be directly connected to the ultracapacitors **60**. Additionally, in some embodiments, the charger **50** may be mounted on the circuit board **36**. The communications adapter **46** is communicatively coupled to a controller **66** of the wind turbine **10**. In some embodiments, the controller **66** is part of the control system of the wind turbine **10**. Accordingly, communications may be passed between the communications adapter **46** and the controller **66**, for example, over the battery communications cable of the wiring harness **26**. In some embodiments, the communications adapter **46** transmits at least one signal comprising information relating to the ultracapacitors **60**. For example, the signal may comprise at least one of a voltage of the ultracapacitors **60**, a charge current of the ultracapacitors **60**, and a temperature of the ultracapacitors **60**.

(63) Additionally, the signal may comprise information relating to a battery test charge state, a current state of the charger (i.e., full charge or charging), or other information relating to a state of the pitch energy module **30**. The battery test charge state may be used to periodically test the ultracapacitors **60**. Here, a test signal is sent to the communications adapter **46** from the controller **66**. Upon receiving the test signal, the pitch energy module **30** transmits a signal comprising a test response signal to the controller **66**. Accordingly, the test response signal may be used to determine whether the pitch energy module **30** is working properly and is responsive to the controller **66**.

(64) In some embodiments, the circuit board **36** or an integrated circuit board of the charger **50** may

be operable to measure various parameters of the pitch energy module **30**. For example, the ESR and the capacitance of the pitch energy module **30** may be measured periodically using a plurality of sensors disposed on the pitch energy module **30** coupled to the circuit board **36**. In some embodiments, a first sensor **68** may be communicatively coupled to the microprocessor **52** operable to measure the ESR of the pitch energy module **30**, as shown. Similarly, a second sensor **70** may be communicatively coupled to the microprocessor **52** operable to measure the capacitance of the pitch energy module **30**. Alternatively, in some embodiments, the sensors **68** and **70** are disposed on the charger **50**. In some embodiments, the ESR and capacitance are measured individually for each ultracapacitor **60**. Alternatively, the overall ESR and capacitance may be measured collectively for the plurality of ultracapacitors **60**. Accordingly, it may be desirable to automatically adjust the voltage set-point of the pitch energy module **30** based on the measured parameters.

(65) In some embodiments, it may be advantageous to automatically adjust the set-point voltage of the pitch energy module **30** based on one of the measured ESR or capacitance such that operators or administrators do not have to manually adjust the set-point voltage of the pitch energy module **30**. It should be understood that prior art voltage adjustments typically require the operator to stop operation of the wind turbine **10** and climb up into the wind turbine **10** to make the adjustment manually. Accordingly, a significant amount of time and energy may be sacrificed to manually adjust the set-point voltage. In some embodiments, the set-point voltage of the pitch energy module **30** may be automatically adjusted by the microprocessor **52** based on signals received from one of the first sensor **68** or the second sensor **70**. It should be understood that in some embodiments, the charger **50** further comprises a microprocessor to perform similar operations as described herein with respect to microprocessor **52**. In some embodiments, both microprocessor **52** and the microprocessor of the charger **50** are included such that processing may be split between the processors.

(66) It should be understood that, as discussed above, the electrical capacitance of the pitch energy module **30** may decrease over time, such that the energy level is resultingly reduced. Accordingly, embodiments are contemplated where the pitch energy module **30** automatically adjusts the voltage set-point based on the measured capacitance such that the energy level within the pitch energy module **30** is maintained over time. For example, the capacitance of the pitch energy module **30** may initially be around 500 Farads with a voltage set-point of 14 Volts. After extended operation, for example, 8 years, the capacitance of the pitch energy module **30** may be measured to be around 480 Farads. Accordingly, the pitch energy module **30** may automatically increase the voltage set-point to 14.5 Volts to accommodate the drop in capacitance. Here, the pitch energy module **30** may automatically adjust the voltage set-point such that a similar amount of energy is provided even after the capacitance of the pitch energy module **30** has decreased.

(67) In some embodiments, information indicative of the signals measured by the first sensor **68** and the second sensor **70** may be communicated to an operator or administrator of the wind turbine **10**, for example, through the communications adapter **46** such that the operator or administrator can monitor the operational parameters of the pitch energy module **30** remotely. Accordingly, the operator or administrator may receive the information indicative of the measured parameters and adjust the voltage set-point of the pitch energy module **30** remotely without having to stop operation of the wind turbine **10** and physically access the wind turbine **10**. Alternatively, in some embodiments, the set-point voltage may be adjusted automatically by the pitch energy module **30** using software executed, for example, on the microprocessor **52**.

(68) Turning now to FIG. 7, a method **700** for using the pitch energy module **30** is depicted relating to some embodiments of the invention. At step **702** at least one ultracapacitor **60** of the pitch energy module **30** is electrically charged using the charger **50**. Here, the power used to charge the ultracapacitors **60** may be received from the external power input **48** or from within the wind turbine **10** such as energy generated by the wind turbine **10**. In some embodiments, the ultracapacitors **60** charge and discharge much faster than the batteries **24** because the electrical

energy from the charger **50** does not need to be converted into chemical energy such as in the batteries **24**.

(69) At step **704** electrical energy is stored within the ultracapacitors **60**. It should be understood that the energy may be stored within the ultracapacitors **60** for an arbitrary amount of time until an emergency pitch energy event occurs or a pitch test is initiated by a user. In some embodiments, each pitch energy module **30** stores about 2 Watt-hours of electrical energy when fully charged. It should be understood that in different embodiments, the energy capacity of the pitch energy module **30** may vary.

(70) At step **706** at least one signal is transmitted to the control system of the wind turbine **10**. For example, the signal may be transmitted to the controller **66** using the communications adapter **46** interfacing with the battery wiring harness **26**. Here, the signal may comprise information associated with the ultracapacitors **60** of the pitch energy module **30** or state information relating to the pitch energy module **30**.

(71) At step **706** the pitch energy module **30** receives a signal from the control system of the wind turbine **10**. Here, the signal may comprise a test signal or a request for a pitch adjustment received from the controller **66** via the communications adapter **46**. If the signal comprises a request for a pitch adjustment, in response to receiving the signal, the pitch energy module **30** will supply energy from the ultracapacitors **60** at step **710**. It should be understood that the pitch energy module **30** may also be configured to supply energy from the ultracapacitors **60** automatically in the case of a power outage where a signal is not received from the controller **66**. The supplied energy may be transmitted to the motors within the wind turbine **10** to adjust the pitch of the blades **18**.

(72) In some embodiments, the pitch may be adjusted manually when a user requests a pitch test. Here, a signal may be sent from the controller **66** to the communications adapter **46** of the pitch energy module **30** requesting a pitch adjustment. Accordingly, energy is supplied from the ultracapacitors **60** and the functionality of the pitch control system is tested.

(73) Turning now to FIG. **8**, a method **800** for installing the pitch energy module **30** in the battery housing **20** is depicted relating to some embodiments of the invention. In some embodiments, the pitch energy module **30** is installed reactively after a battery **24** has been deemed faulty or has exceeded its useful lifetime. Alternatively, the pitch energy module **30** may be installed proactively, before the battery **24** has exceeded its useful lifetime to increase the efficiency of the pitch energy system.

(74) At step **802** the battery wiring harness **26** is disconnected from the battery **24**. Here, a plurality of cables of the battery wiring harness **26** may be unplugged and removed from the battery **24**. For example, the battery wiring harness **26** may comprise cables disconnected from a positive terminal of the battery, a negative terminal of the battery, a communications adapter of the battery, and an external power input of the battery. At step **804** the battery **24** is removed from the battery housing **20**. Here, the battery **24** may be removed by removing bolts disposed within mounting brackets of the battery **24** configured to secure the battery **24** within the frame **22** of the battery housing **20**.

(75) At step **806** the pitch energy module **30** is mounted within the battery housing **20** in place of the battery **24**. Here, the bolts are disposed within the mounting brackets **34** of the pitch energy module housing **32** to secure the pitch energy module **30** to the frame **22** of the battery housing **20**. In some embodiments, as discussed above the mounting brackets **34** may comprise slotted holes such that the pitch energy module **30** is adjustably mounted within the battery housing **20**. Accordingly, space within the battery housing **20** may be optimized by slightly adjusting the mounting position of the pitch energy module **30**.

(76) At step **808** the battery wiring harness **26** is connected to the pitch energy module **30**. Accordingly, a plurality of cables of the battery wiring harness **26** may be plugged into the pitch energy module **30**. For example, cables of the battery wiring harness **26** may be connected to the positive terminal **40**, the negative terminal **42**, the communications adapter **46**, and the external power input **48**. Accordingly, the pitch energy module **30** is installed within the battery housing **20**

without further modifications to the battery housing **20** or the battery wiring harness **26**, such as additional communication cables or mounting structure or hardware. Here, the same battery wiring harness **26** from the battery **24** is used to connect the pitch energy module **30**. In some such embodiments, it may be desirable to use the same battery wiring harness **26** and mounting mechanism such that the installation cost associated with the pitch energy module **30** is reduced. Further, the pitch energy module **30** may be installed without changing the control system of the wind turbine **10** because the pitch energy module **30** uses the same battery wiring harness and communicates with the control system of the wind turbine **10** using the same cables.

(77) Features described above as well as those claimed below may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible, non-limiting combinations:

(78) (A1) A pitch energy module for replacing at least one battery within an electric pitch control system of a wind turbine, the at least one battery mounted in the wind turbine in a battery housing and operably coupled with the wind turbine's control system via at least a battery wiring harness. The pitch energy module includes a pitch energy module housing sized for mounting in the battery housing upon replacement of the at least one battery with the pitch energy module. One or more ultracapacitors are mounted within the pitch energy module housing, the one or more ultracapacitors configured to store electrical energy for a wind turbine emergency pitch event. The pitch energy module further includes a microprocessor mounted on or in the pitch energy module housing, the microprocessor processing a first set of information associated with the one or more ultracapacitors. A non-volatile memory is communicatively coupled with the microprocessor for storing a second set of information associated with the pitch energy module. A communications adapter, mounted on or in the pitch energy module housing is communicatively coupled with the microprocessor, the communications adapter configured to interface with a battery communications cable of the battery wiring harness to thereby transfer at least one signal comprising the first set of information associated with the one or more ultracapacitors to the control system of the wind turbine. The pitch energy module further includes a charger mounted on or in the pitch energy module housing for electrically charging the one or more ultracapacitors.

(79) (A2) For the pitch energy module denoted as (A1), the module may further include a positive terminal electrically coupled to the one or more ultracapacitors, the positive terminal comprising a first bolt fastener for increasing an equivalent series resistance of the pitch energy module, and a negative terminal electrically coupled to the one or more ultracapacitors, the negative terminal comprising a second bolt fastener for increasing the equivalent series resistance of the pitch energy module.

(80) (A3) For the pitch energy module denoted as any of (A1) or (A2), the module may be electrically connected in series with a second pitch energy module that is disposed within the battery housing of the wind turbine.

(81) (A4) For the pitch energy module denoted as any of (A1) through (A3), the pitch energy module may be electrically connected in series with a battery that is disposed within the battery housing of the wind turbine.

(82) (A5) For the pitch energy module denoted as any of (A1) through (A4), the pitch energy module further including an alternating current power input mounted on or in the pitch energy module housing for receiving electrical energy from an external power source via a power cable of the battery wiring harness to temporarily power the pitch energy module during a grid event.

(83) (A6) For the pitch energy module denoted as any of (A1) through (A5), the charger may be selectably mounted on one of a front facing side of the pitch energy module housing or a top side of the pitch energy module housing based on a configuration of the wind turbine.

(84) (A7) For the pitch energy module denoted as any of (A1) through (A6), the pitch energy module may further include an operating voltage of the one or more ultracapacitors that is configurable by a user to thereby increase a lifetime of the one or more ultracapacitors.

(85) (A8) For the pitch energy module denoted as any of (A1) through (A7), the pitch energy module may further include a circuit board mounted on or in the pitch energy module housing, the circuit board is communicatively coupled to the microprocessor and the non-volatile memory. The circuit board may include a high current circuit trace for transmitting power by electrically connecting the one or more ultracapacitors in series.

(86) (A9) For the pitch energy module denoted as any of (A1) through (A8), the pitch energy module may further include a protective fiberglass cover mounted to the pitch energy module housing and disposed over the circuit board for protecting the circuit board.

(87) (A10) For the pitch energy module denoted as any of (A1) through (A9), the pitch energy module may be configured to be electrically connected to the battery wiring harness without depinning the battery wiring harness.

(88) (B1) A pitch energy module for replacing a battery within an electric pitch control system of a wind turbine, the battery mounted in the wind turbine in a battery housing and operably coupled with the wind turbine's control system via at least a battery wiring harness. The pitch energy module including a pitch energy module housing sized for mounting in the battery housing upon replacement of the battery with the pitch energy module. One or more ultracapacitors are mounted within the pitch energy module housing, the one or more ultracapacitors configured to store electrical energy for a wind turbine emergency pitch event. A microprocessor is mounted on or in the pitch energy module housing, the microprocessor for processing a first set of information associated with the one or more ultracapacitors. A non-volatile memory is communicatively coupled with the microprocessor for storing a second set of information associated with the pitch energy module. A positive terminal is electrically coupled to the one or more ultracapacitors, the positive terminal comprising a first bolt fastener. A negative terminal is electrically coupled to the one or more ultracapacitors, the negative terminal comprising a second bolt fastener. The battery being replaced comprises a first equivalent series resistance. The first bolt fastener and the second bolt fastener achieve a second equivalent series resistance for the pitch energy module that is higher than the first equivalent series resistance of the battery. A communications adapter is mounted on or in the pitch energy module housing and is communicatively coupled with the microprocessor. The communications adapter is configured to interface with a battery communications cable of the battery wiring harness to thereby transfer at least one signal comprising the first set of information associated with the one or more ultracapacitors to the control system of the wind turbine. The pitch energy module further includes a charger mounted on or in the pitch energy module housing for electrically charging the one or more ultracapacitors.

(89) (B2) For the pitch energy module denoted as (B1), the pitch energy module may further include a plurality of mounting brackets including slots for adjustably securing the pitch energy module within a battery mounting position in the battery housing of the wind turbine.

(90) (B3) For the pitch energy module denoted as any of (B1) or (B2), the pitch energy module may be disposed within a rotor of the wind turbine with at least one battery, and each of the pitch energy module and the at least one battery may supply electrical power during a wind turbine emergency pitch event.

(91) (B4) For the pitch energy module denoted as any of (B1) through (B3), the charger may be selectably mounted on one of a front facing side of the pitch energy module housing or a top side of the pitch energy module housing based on a configuration of the wind turbine, and the charger may be held at a floating voltage potential of the pitch energy module.

(92) (C1) A pitch energy system for replacing at least one battery within an electric pitch control system of a wind turbine, the at least one battery mounted in the wind turbine in a battery housing and operably coupled with the wind turbine's control system via at least a battery wiring harness. The pitch energy system includes a plurality of pitch energy modules electrically connected in series. Each of the pitch energy modules includes a pitch energy module housing sized for mounting in the battery housing upon replacement of the at least one battery with the pitch energy

module. One or more ultracapacitors are mounted within the pitch energy module housing, the one or more ultracapacitors configured to store electrical energy for a wind turbine emergency pitch event. The pitch energy module further includes a microprocessor mounted on or in the pitch energy module housing, the microprocessor processing a first set of information associated with the one or more ultracapacitors. A non-volatile memory is communicatively coupled with the microprocessor for storing a second set of information associated with the pitch energy module. A communications adapter, mounted on or in the pitch energy module housing is communicatively coupled with the microprocessor, the communications adapter configured to interface with a battery communications cable of the battery wiring harness to thereby transfer at least one signal comprising the first set of information associated with the one or more ultracapacitors to the control system of the wind turbine. The pitch energy module further includes a charger mounted on or in the pitch energy module housing for electrically charging the one or more ultracapacitors.

(93) (C2) For the pitch energy system denoted as (C1), the pitch energy system may further include a battery electrically connected in series with the plurality of pitch energy modules, the battery configured to store electrical energy for the wind turbine emergency pitch event.

(94) (C3) For the pitch system module denoted as any of (C1) or (C2), during the wind turbine emergency pitch event, a first portion of electrical energy for the wind turbine emergency pitch event may be supplied by the plurality of pitch energy modules and a second portion of electrical energy for the wind turbine emergency pitch event may be supplied by the battery.

(95) (C4) For the pitch energy system denoted as any of (C1) through (C3), each of the pitch energy modules may further include a circuit board mounted on or in the pitch energy module housing. The circuit board may be communicatively coupled to the microprocessor and the non-volatile memory. The circuit board may further include a high current circuit trace for transmitting power by electrically connecting the one or more ultracapacitors in series.

(96) (C5) For the pitch energy system denoted as any of (C1) through (C4), the second set of information stored on the non-volatile memory may further include at least one of a number of wind turbine emergency pitch events, a number of full charges of the respective pitch energy module, and a number of days of operation of the respective pitch energy module.

(97) (C6) For the pitch energy system denoted as any of (C1) through (C5), the charger of each of the plurality of pitch energy modules may be selectably mounted on one of a front facing side of the respective pitch energy module housing or a top side of the respective pitch energy module housing based on a configuration of the wind turbine.

(98) Although the invention has been described with reference to the embodiments illustrated in the attached drawing figures, it is noted that equivalents may be employed and substitutions made herein without departing from the scope of the invention as recited in the claims.

Claims

1. A pitch energy module for replacing at least one battery within an electric pitch control system of a wind turbine, the pitch energy module comprising: a pitch energy module housing sized for mounting in a battery housing of the wind turbine; one or more capacitors mounted within the pitch energy module housing, the one or more capacitors configured to store electrical energy for a wind turbine pitch event; and a control system programmed to adjust a voltage set-point of the pitch energy module based on at least one measured electrical parameter of the pitch energy module to thereby increase a lifetime of the one or more capacitors of the pitch energy module.
2. The pitch energy module of claim 1, further comprising: at least one sensor operable to measure the at least one measured electrical parameter of the pitch energy module.
3. The pitch energy module of claim 2, wherein the at least one measured electrical parameter of the pitch energy module includes an equivalent series resistance of the pitch energy module.
4. The pitch energy module of claim 3, wherein the at least one measured electrical parameter of

the pitch energy module further includes a capacitance of the pitch energy module.

5. The pitch energy module of claim 1, further comprising: one or more electrical terminals disposed external to the pitch energy module housing.

6. The pitch energy module of claim 5, further comprising: one or more bolt fasteners operable to secure cables to the one or more electrical terminals.

7. The pitch energy module of claim 6, wherein an equivalent series resistance of the pitch energy module is about 16 milliohms.

8. The pitch energy module of claim 1, wherein the one or more capacitors comprise an ultracapacitor.

9. A control system for an energy module of a wind turbine, the energy module for replacing at least one battery within an electric pitch control system of the wind turbine and comprising an energy module housing sized for mounting in a battery housing of the wind turbine and one or more capacitors mounted within the energy module housing, the one or more capacitors configured to store electrical energy for a wind turbine pitch event, the control system comprising: at least one processor communicatively coupled to the one or more capacitors; at least one sensor operable to measure at least one measured electrical parameter of the energy module; and one or more non-transitory computer-readable media that store computer-executable instructions that, when executed by the at least one processor, automatically adjust a voltage set-point of the energy module based on the at least one measured electrical parameter of the energy module to thereby increase a lifetime of the one or more capacitors of the energy module.

10. The control system of claim 9, wherein the one or more non-transitory computer-readable media that store computer-executable instructions that, when executed by the at least one processor, are further operable to store a signal including information indicative of the at least one measured electrical parameter.

11. The control system of claim 9, wherein the one or more capacitors comprises at least one hybrid capacitor comprising an electrostatic storage portion and an electrochemical storage portion.

12. The control system of claim 9, wherein an equivalent series resistance of the energy module is about 16 milliohms.

13. The control system of claim 9, further comprising: a communications adapter communicatively coupled to a control system of the wind turbine.

14. The control system of claim 13, wherein the communications adapter is operable to transmit a signal indicative of the at least one measured electrical parameter of the energy module.

15. An energy module for replacing at least one battery within an energy storage system of a wind turbine, the energy module comprising: an energy module housing sized for mounting in a battery housing of the wind turbine; one or more capacitors mounted within the energy module housing, the one or more capacitors configured to store electrical energy for a wind turbine pitch event; at least one sensor operable to measure at least one measured parameter of the energy module; and a control system programmed to adjust a voltage set-point of the energy module based on at least one measured parameter of the energy module to thereby increase a lifetime of the one or more capacitors of the energy module.

16. The energy module of claim 15, further comprising: a circuit board comprising a circuit trace coupled to the one or more capacitors.

17. The energy module of claim 15, wherein the energy module is operable to be arranged electrically in series with at least one battery within the energy storage system of the wind turbine.

18. The energy module of claim 15, wherein the at least one measured parameter includes a temperature associated with the one or more capacitors.

19. The energy module of claim 15, further comprising: a charger coupled to the one or more capacitors operable to charge the one or more capacitors.

20. The energy module of claim 19, wherein the charger is mounted external to the energy module housing.

